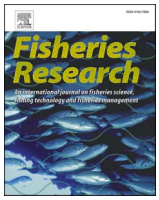
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Model-based indices of juvenile Pacific salmon abundance highlight species-specific seasonal distributions and impacts of changes to survey design

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ABSTRACT

Changes in survey implementation can bias traditional design-based indices of abundance, particularly for migratory species where local abundance can vary at short temporal scales. Spatiotemporal model-based estimators provide a flexible alternative that can better account for changes in the timing and location of surveys. Here we develop a geostatistical model, a generalized linear mixed effects model with Gaussian Markov random fields, that is sufficiently flexible to account for changes in survey design, as well as seasonal and interannual variability in abundance and spatial distribution. We then apply this model to a suite of migratory species—juvenile Pacific salmon—with survey data collected from southern British Columbia's continental shelf over the previous 25 years. Juvenile Pacific salmon showed species-specific spatial distributions with considerable seasonal variability. While the location of species-specific hot spots varied among years, seasonal variation in distributions were greater than interannual variation. We found that a shift from standard transects and opportunistic sampling to a random stratified survey design resulted in lower estimates of abundance for four of five juvenile Pacific salmon species; however, these differences were not significant after accounting for other sampling attributes. Trends in abundance for several species shifted when models included survey and diel effects. Our approach provides a means of deriving indices of abundance for migratory species from surveys with variable effort in time and space. Ultimately such indices can be used to better understand the ecological mechanisms regulating productivity by linking distribution and abundance to environmental drivers.

1. Introduction

Estimates of relative abundance or biomass are critical stock assessment inputs that guide fisheries management decisions. While catch per unit effort is commonly used to represent relative abundance in stock assessments, such indices can be biased by a range of confounding variables (Hilborn and Walters, 1992; Harley et al., 2001; Maunder and Punt, 2004). In the case of fisheries-dependent data, effort may be concentrated in areas or time periods of high biomass, rather than systematically distributed. Using indices of abundance in fished areas to predict abundance in unfished areas may result in bias and, ultimately, mis-specified fisheries reference points. Fisheries-independent surveys, standardized to have consistent sampling methods and spatio-temporal coverage, can address some of these shortcomings (Maunder and Piner, 2015). Yet fisheries-independent surveys, which are costly and logistically challenging to operate, are often impacted by changes in survey design or sampling effort that can result in similarly biased outcomes (ICES, 2023).

Historically, changes in survey design have often been addressed in an ad hoc manner, for example, by excluding intermittently sampled strata from analyses, imputing missing values, or applying expansion factors to adjust for variable effort (ICES, 2023). Model-based estimates of abundance, unlike design-based approaches, assume that abundance can be associated with an underlying statistical distribution (Sarndal, 1978; Pennington, 1983; Aubry and Francesiaz, 2022). Model-based approaches can account for changes in vessel characteristics or sample timing using covariates (Maunder and Punt, 2004). The underlying spatial structure associated with sampling can be addressed in multiple ways. For example, generalized additive models, commonly implemented using the mgcv package (Wood, 2011), can incorporate spatial processes via two-dimensional splines that model non-linear effects of location (Kammann and Wand, 2003). Alternatively, latent spatial

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relationships can be represented by Gaussian random fields where spatial patterns are assumed to follow a multivariate normal distribution with some covariance function (Rue and Held, 2005; Diggle and Ribeiro, 2007; Lindgren et al., 2011). Packages such as inlabru (Bachl et al., 2019), VAST (Thorson, 2019), and sdmTMB (Anderson et al., 2024) use a stochastic partial differential equation (SPDE) approach to approximate Gaussian random fields with Gaussian Markov random fields and link latent spatial processes to generalized linear models (GLMs) that are widely used in ecology. Such species distribution models have informed a wide range of stock, climate, and ecosystem assessments (e.g., Thorson, 2019; Anderson et al., 2019b).

Unlike stock assessments for many marine fishes, indices of abundance derived from marine surveys are rarely incorporated directly into Pacific salmon stock assessments. Instead, stock status is evaluated based on estimates of adult catch and escapement (the number of individuals estimated to arrive on spawning grounds). However, Pacific salmon recruitment dynamics are hypothesized to be regulated by survival during early marine periods (Beamish and Mahnken, 2001). Therefore estimates of abundance immediately after this recruitment bottleneck may function as early signals of recruitment (e.g., Beamish et al., 2013; Murphy et al., 2017) and have been used to forecast abundance (Burke et al., 2013; Howard et al., 2020; Miller et al., 2022). More generally, spatially and temporally explicit estimates of abundance can be used to identify factors that moderate salmon productivity and guide ecosystem-based management (Wells et al., 2016,2020; Farley et al., 2024).

Fisheries and Oceans Canada (DFO) has operated near-surface trawl surveys on the continental shelf of British Columbia since 1998. Initial survey objectives were to identify migration routes of juvenile Pacific salmon so that mechanisms regulating juvenile survival could be better understood (Welch et al., 2000). Researchers paired catch data with coded wire tag recoveries and genetic stock identification techniques to identify both species- and stock-specific migratory behaviours (Trudel et al., 2009; Tucker et al., 2009,2011). Analyses of biological samples provided novel information on juvenile salmon diets (Brodeur et al., 2007; Hertz et al., 2015), growth and physiology (Trudel et al., 2007; Hertz et al., 2017; Freshwater et al., 2016a), migratory behaviours (Freshwater et al., 2016b), and predation (Tucker et al., 2016).

Data collected from DFO juvenile salmon surveys have been used intermittently and informally to guide management decisions. For example, the condition of juvenile coho salmon captured on the west-coast of Vancouver Island has been used as an indicator of future returns (Trudel et al., 2016), while the abundance and condition of Fraser River sockeye salmon informs post-season qualitative reviews (MacDonald et al., 2019). However, changes in survey objectives over time, inconsistent funding, and constraints in vessel availability have resulted in considerable interannual variability in the timing and location of DFO's sampling effort along the British Columbia coastline.

Additionally, summer surveys were redesigned in 2017 resulting in a shift from standard transects and opportunistic sampling (largely during daylight hours only) to paired diurnal/nocturnal sampling of stations randomly selected in depth-defined strata (King et al., 2019). The survey is now intended to sample both juvenile salmon and pelagic forage fish (e.g., Pacific herring *Clupea pallasii*, Pacific sardine *Sardinops sagax*), which were previously indexed by separate surveys. The current survey also enumerates and samples the pelagic ecosystem more broadly, including oceanographic sampling of primary productivity and zooplankton, acoustic surveys of euphausiids, and trawl sampling of potential competitors and predators (e.g., jellyfish, mackerels, Pacific spiny dogfish *Squalus suckleyi*, Pacific hake *Merluccius productus*). Ultimately changes in survey design were intended to allow DFO to better meet various ecosystem-based management objectives, such as accounting for interactions among trophic levels and changing environmental conditions, as well as providing species-specific indices of abundance for assessment and maps of species distributions for spatial planning. Despite the long-term benefits of the new survey design, it introduced confounding effects that were necessary to quantify before drawing inference from the integrated time series.

Developing robust indices of juvenile salmon marine abundance is challenging due to several ecological characteristics. Like many pelagic forage fishes (Boyd et al., 2015), juvenile Pacific salmon are patchily distributed, resulting in spatial heterogeneity that requires substantial survey effort to adequately estimate spatial extent and abundance. Additionally juvenile salmon migrate away from ocean entry locations during their first year of marine residence, with migration phenologies and routes differing among species (Fisher et al., 2007). Thus relatively minor changes in the amount, timing, or location of survey effort may reduce the precision of abundance indices. Spatiotemporal models provide a means of accounting for changes in survey design, while simultaneously quantifying juvenile salmon distribution and abundance at seasonal and interannual scales.

To generate indices of species-specific abundance, we developed spatiotemporal models that accounted for spatial and spatiotemporal variability in catch rates via Gaussian Markov random fields. Our approach is similar to models used to estimate juvenile chum salmon *Oncorhynchus keta* abundance and distributions in the Bering Sea using VAST (Farley et al., 2024); however, our approach includes several novel components. First, our model emphasizes sampling effects including season, headrope depth, diurnal versus nocturnal sampling, and overall survey design. Second, we model season-specific Gaussian Markov random fields simultaneously, allowing us to leverage temporal covariation within a year to account for gaps in sampling among years. Third, we applied the same modelling framework to five species of juvenile Pacific salmon, allowing for robust comparisons. Our results highlight differences among species in marine distributions and provide an index of changes in abundance over 25 years of fisheries-independent surveys along British Columbia's continental shelf.

2. Methods

2.1. Sample collection

Our analysis focused on five Pacific salmon species: Chinook *O. tshawytscha*, chum, coho *O. kisutch*, pink *O. gorbuscha*, and sockeye salmon *O. nerka*. Pacific salmon were collected by midwater trawl using a variety of commercial and research vessels between 1998 and 2022. Samples were collected throughout northern Washington, British Columbia, and southeast Alaska, with the majority of effort focused on the continental shelf. Given that the northern- and southern-most portions of the survey area were rarely sampled, we constrained our analysis to an area extending approximately from Haida Gwaii in the north to Grays Harbor in the south (Fig. 1). Sampling occurred intermittently in most months, but was concentrated in March, June, July, and October (Fig. 2). For the purposes of our analysis sampling events were binned by season, defined as spring (February–April), summer (May–August) and fall (September–December). No sampling occurred in January.

Sampling events differed in headrope depth (Figure S1, Table S1), diel sampling period (day/night; Figures S2, S3), and volume swept (the product of net opening size, tow duration, and vessel speed; Figure S4, Table S1). Net mouth openings (width and height) were estimated from vessel net mensuration sensors. The net that was used for sampling changed in 2018 from a Cantrawl 250 to a LFS 7742 mid-water trawl net. Since the latter net was designed to be comparable to the former and a net comparison study indicated catchability for juvenile salmon was not significantly different between the two (Anderson et al., 2019a), we did not directly account for the change in gear within our analysis. Prior to 2017, sampling occurred as part of Fisheries and Oceans Canada's (DFO) High Seas Salmon (HSS) program and sampling stations were selected to emphasize cross shelf transects and coastal inlets. Target headrope depth for midwater trawl operations during HSS surveys was typically at the surface (i.e., zero meters) or 15 meters and tow duration was typically 30 minutes; however, deeper headrope depths (up to 60

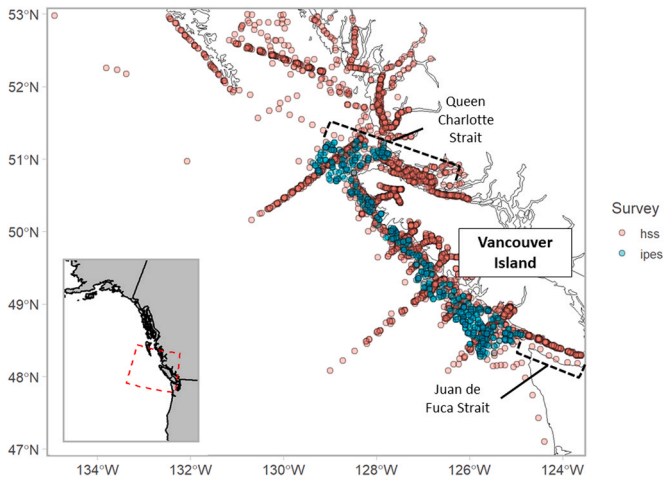


Fig. 1. Spatial extent of station locations included in model fitting. High Seas Salmon (HSS) sampling locations are shown in red. Integrated Pelagic Ecosystem Survey (IPES) locations, constrained to a predefined grid and fished during summer months after 2016, are shown in blue. Projection is WGS84 and shapefiles for coastlines are from the rnaturalearth package version 0.1.0 (South, 2017).

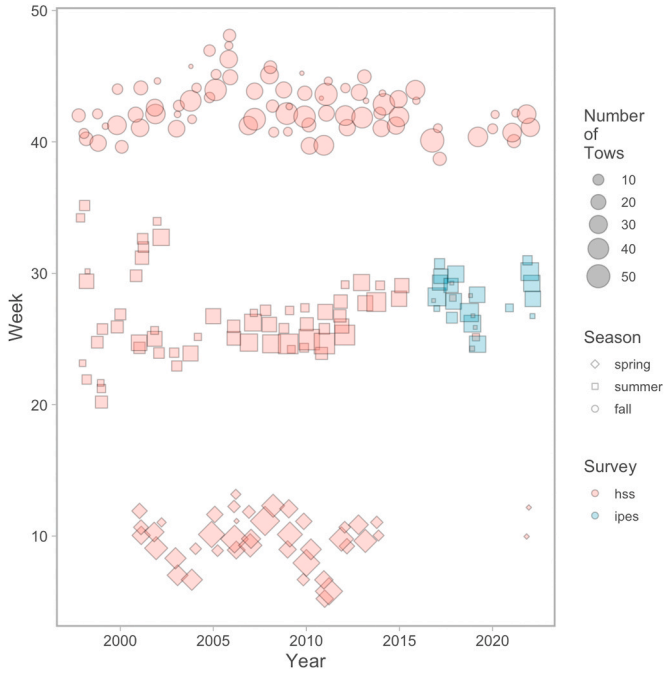


Fig. 2. Seasonal and annual survey coverage. Area of symbols corresponds to the number of completed sets within a given survey. Color of symbols corresponds to survey design. Shape of symbols corresponds to seasonal categories used in model fitting.

meters) and longer tows also occurred.

In 2017, the structure of DFO's summer surveys changed to focus on pelagic ecosystems more broadly. The contemporary Integrated Pelagic Ecosystem Survey (IPES) uses a random design where stations are stratified by bioregion and depth, and each station is sampled once during daylight hours and once after dark (King et al., 2019). The IPES survey domain is constrained to continental shelf waters off northern and western Vancouver Island from Queen Charlotte Strait to Juan de Fuca Strait. However due to limited vessel availability, surveys during fall months have not been redesigned and since 2017 have continued to sample historical stations. In both summer and fall surveys, only surface and 15 m headrope depths have been fished since 2017.

Regardless of survey design, Pacific salmon were landed, sorted by species and life stage, with juveniles defined as individuals likely to have spent less than one year at sea, and enumerated. Life-stage assignments were determined based on length frequency distributions, which differed by season and species. For a subset of stations ($\approx$ 2%), the total catch was subsampled and enumerated before being expanded to represent the total catch because large or species-rich catches prevented the entire tow from being sampled individually.

2.2. Modeling framework

We used species-specific generalized linear models with spatial and spatiotemporal Gaussian Markov random fields to separate the effects of survey design (including variable effort seasonally and spatially) from long-term patterns in catch per unit effort, a proxy for abundance. This model structure allowed us to estimate the fixed effect of distinct sampling variables, as well as residual variability associated with spatially correlated latent processes. Juvenile Pacific salmon commonly exhibit stock-specific differences in migration rate and marine distributions, which result in seasonal changes in spatial distribution within the survey domain (Fisher et al., 2007; Tucker et al., 2011). We accounted for this seasonal variability by estimating seasonal effects on mean catch rate, as well as trends in catch rate among years. We also allowed latent spatial patterns associated with interannual changes in distribution to vary among seasons. Although the abundance of different Pacific salmon species may be correlated, we fit each species' model separately given differences in juvenile migration patterns and ecology (Groot and Margolis, 1991; Fisher et al., 2007). We also note that explicitly modeling covariance does not necessarily improve the precision of abundance estimates from fisheries-independent surveys for well-sampled species (Thorson and Barnett, 2017).

Since species catches were based on counts of individuals (not biomass), and zero catches were common (Table S1), we modelled catch using a negative binomial ('NB2', Hilbe, 2011) distribution and a log link:

$$c_{i,t} \sim \text{NegBin}\left(\mu_{i,t}, \phi\right), \tag{1}$$

$$\log\left(\mu_{i,s,t}\right) = \alpha_s + \alpha_f + \alpha_d + \beta_h h_{i,t} + \beta_g g_{i,t} + \log\left(p_{i,t}\right) + \omega_{i,s} + \delta_{i,s,t}, \tag{2}$$

$$\boldsymbol{\omega}_s \sim \text{MVN}\left(0, \boldsymbol{\Sigma}_\omega\right), \tag{3}$$

$$\boldsymbol{\delta}_{s,t=1} \sim \text{MVN}\left(\boldsymbol{0}, \boldsymbol{\Sigma}_\epsilon\right), \tag{4}$$

$$\boldsymbol{\delta}_{s,t>1} = \boldsymbol{\delta}_{s,t-1} + \boldsymbol{\epsilon}_{s,t-1},\ \boldsymbol{\epsilon}_{s,t-1} \sim \text{MVN}\left(\boldsymbol{0}, \boldsymbol{\Sigma}_\epsilon\right), \tag{5}$$

where $c_{i,t}$ represents catch at location i and year t, μ represents the mean catch, and ϕ represents the inverse dispersion parameter (variance $= \mu + \mu^2/\phi$). The α parameters represent categorical mean effects season s, survey design f (IPES or HSS), and diel period d (day or night), each coded as factors. The β parameters represent linear effects of headrope depth h and distance to coastline g. We expected catch rates to decline with increasing headrope depth and distance to coastline because juvenile salmon are generally surface- and continental shelf-oriented (Fisher et al., 2007; Quinn, 2018). We included the natural logarithm of the estimated volume swept by the trawl net (p) as an effort offset (McCullagh and Nelder, 1989).

We accounted for latent seasonal differences in spatial distribution, as well as interannual variation in spatial distribution that differed among seasons (i.e., season-specific spatial and spatiotemporal random fields). We included Gaussian Markov random fields constrained by Matérn covariance functions for each season with $\boldsymbol{\omega}_s$ denoting a vector for a season and $\omega_{i,s}$ a value at a point in space for a season. We also modeled interannual changes in distribution (i.e. spatiotemporal variation) using season-specific random fields following a random walk ($\boldsymbol{\epsilon}_{s,t}$

and $\delta_{i,s,t}$). Spatial and spatiotemporal random fields were fit with shared marginal standard deviation (covariance matrices $\boldsymbol{\Sigma}_\omega$ and $\boldsymbol{\Sigma}_\epsilon$). All random fields shared a common κ parameter defining the Matérn decorrelation rate with distance. Spatial correlations were modeled as anisotropic (i.e., directionally dependent) (Haskard, 2007; Thorson et al., 2015). While it would be more ecologically accurate to model season-specific random fields with independent spatial decorrelation rates (κ), preliminary models that estimated separate spatial and spatiotemporal κ parameters failed to converge. While the sdmTMB package does not currently allow κ to vary among categories (here, seasons), it is likely that such a formulation would also lead to convergence issues.

We modeled spatial random fields using a triangulated mesh (Lindgren et al., 2011) with a minimum distance between vertices of 30 km and a maximum edge distance of 500 km, resulting in a mesh with 169 vertices. We then used the estimated values at each knot and bilinear interpolation to approximate a continuous fields that represent both spatial and spatiotemporal effects (Lindgren et al., 2011).

We also considered a model structure that included temporal fixed effects for each species. Temporal fixed effects were represented by α_x where x represented species-specific temporal covariates. In the case of Chinook, chum and coho salmon, x was equivalent to t, resulting in year-specific intercepts. Such a model failed to converge for pink and sockeye salmon, in which case we instead estimated cycle line-specific intercepts to account for the present of dominant and subdominant cohorts that occur at regular intervals in southern British Columbia waters. In the case of pink salmon, cycles are biennial, while sockeye salmon cycles occur at four-year intervals (Groot and Margolis, 1991). Temporal fixed effects models allow mean abundance to vary among factor levels (year or cycle line), while the original model used spatiotemporal random fields alone to estimate interannual changes in abundance. Marginal AIC (Akaike information criterion) values in a preliminary analysis indicated the most parsimonious model for pink and sockeye salmon included fixed effects for cycle line, while in the remaining species there was greater support for including season-specific spatiotemporal random fields alone (ΔAIC $>$ 8). In other words, no models included year-specific intercepts.

We used conditional predictions to evaluate the relative impact of individual fixed effects on each species, while holding other covariates constant at reference values. We explored spatial patterns in each species' catch rate by projecting model predictions onto a 1 $\times$ 1 km grid (UTM 9 projection). Finally, we generated species-specific area-weighted indices of abundance by projecting spatial predictions onto a similar grid, summing estimated catch in each grid cell (e.g., Thorson et al., 2015), and applying a generic bias-correction (Thorson et al., 2016). We controlled for variation in survey effort by generating conditional predictions with survey effects fixed at the HSS estimate (since the majority of sampling events fell into this category), with a zero meter headrope depth, diurnal sampling, and the mean observed volume swept. We generated indices for both summer and fall surveys using season-specific spatial domains that represented recent and anticipated future survey effort. For summer predictions, this was the IPES survey domain, whereas fall predictions were constrained to a smaller domain in northwest Vancouver Island and Queen Charlotte Strait where the majority of sampling effort has occurred in recent years (Figure S5).

Finally, we evaluated the impact of accounting for survey effects. We estimated effect sizes for day/night and survey design parameters. We compared models that excluded day/night and survey effects (i.e., did not include α_d and α_f) to the full model (Equation (2)) using marginal AIC. Preliminary efforts to use cross-validation to evaluate the impact of these parameters were not successful due to relatively little sampling in recent years, which lead to model convergence issues for several species. To evaluate survey effects on inferred status, we also generated indices of abundance from each model. We fit a linear model to the abundance index, for each species and model, from summer surveys over the previous 12 years (approximately three generations for most species). We then compared the slope estimate (i.e. the year effect) for each species and model. Fall surveys were excluded from this portion of the analysis since the design of that survey has remained consistent. We partially accounted for uncertainty in abundance indices by weighting data within each linear model by the inverse of the standard error of the estimate. We note that this method fails to fully propagate uncertainty, but is consistent with, for example, equivalent Committee on the Status of Endangered Wildlife in Canada analyses, which treat abundance as point estimates (COSEWIC, 2015).

We fit models using the sdmTMB package version 0.3.1 ('mvrfrw' branch) (Anderson et al., 2024), which combines automatic differentiation and the Laplace approximation via the TMB package version 1.9.10 (Kristensen et al., 2016) with input matrices for the SPDE approximation to Gaussian random fields from the INLA package version 22.6.25 (Rue et al., 2009) and minimizes the negative marginal log likelihood of the model with the non-linear minimizer `nlminb()` in R (R Core Team, 2021). We fit models in R version 4.2.1 (R Core Team, 2021).

We confirmed that the non-linear optimizer had converged by checking that the Hessian matrix was positive definite and the maximum absolute gradient for all fixed effects was $<$ 0.005. We calculated randomized quantile residuals with fixed-effects fixed at their maximum likelihood estimates (MLEs) and random effects sampled with Markov chain Monte Carlo (MCMC) (*sensu* Rufener et al., 2021) to evaluate residual diagnostics independent of the Laplace approximation.

Spatiotemporal models with latent GMRFs, such as the ones applied here, are complex and important to validate. We used a self-simulation exercise to verify that our models were self consistent. Briefly, we used estimated parameters from fitted models (fixed effects at their MLEs and random effects sampled with MCMC) to simulate 100 datasets for each species with new observations drawn from the estimated observation error. We then refit each species' model to the simulated datasets to determine whether parameters could be reliably recovered. We also generated season-specific abundance indices from 100 of the refit models, which we compared to the original abundance index estimated for each species.

3. Results

Juvenile Pacific salmon catch rates (individuals per km^3) covaried with several survey characteristics. Abundance was lowest for all species during the spring survey period. Differences in abundance between the summer and fall periods were more modest; however, mean abundance during summer was typically greatest, particularly for sockeye salmon (Fig. 3). Abundance was significantly lower during nocturnal tows for chum, coho and sockeye salmon, while there was no significant diel effect for Chinook or pink salmon (Fig. 3). The transition from the HSS to IPES survey design was correlated with reduced catch rates for all species except pink salmon; however, these differences were not statistically significant for any species (Fig. 3).

Mean abundance declined with increasing headrope depth and was negligible below 20 m for four of the five juvenile salmon species (Fig. 4). Chinook salmon were the only exception to this pattern—abundance declined with depth, but near linearly (and with large uncertainty), resulting in moderate predicted mean abundance even at headrope depths greater than 60 m (Fig. 4). Abundance declined with distance from shore for all species, with catches declining to $<$ 50% of their maximum value at approximately 15 km from shore (Fig. 4).

Juvenile Pacific salmon spatial distributions varied among species and seasons (Figs. 5, S6, S7). Chinook salmon had diffuse distributions along the shelf during the summer and were concentrated in west coast inlets during the fall, while chum salmon showed the opposite pattern (Fig. 5). Like chum salmon, coho salmon were distributed inshore during the summer, but median abundance shifted to southwest near Juan de Fuca Strait during fall (Fig. 5). The median abundance of pink and sockeye salmon was typically greatest in Queen Charlotte Strait (Figs. 5,

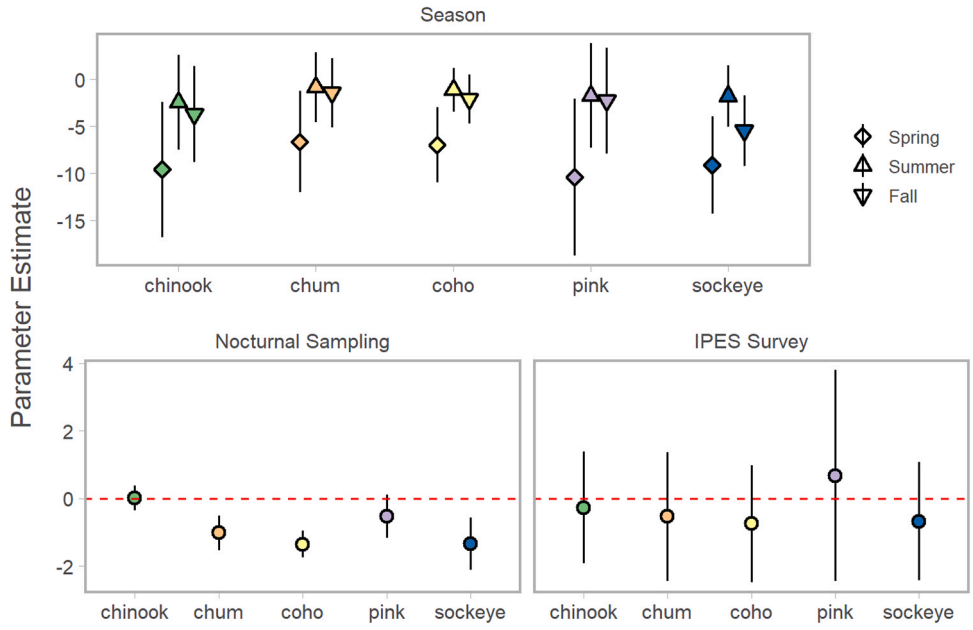


Fig. 3. Species-specific parameter estimates for categorical fixed effects on juvenile salmon abundance. Seasonal effects (top panel) are represented by intercepts, while diel period (bottom left) and survey design (bottom right) are estimated relative to reference values (daytime sampling and the HSS survey design, respectively). Points represent mean estimates in link space and whiskers 95% confidence intervals.

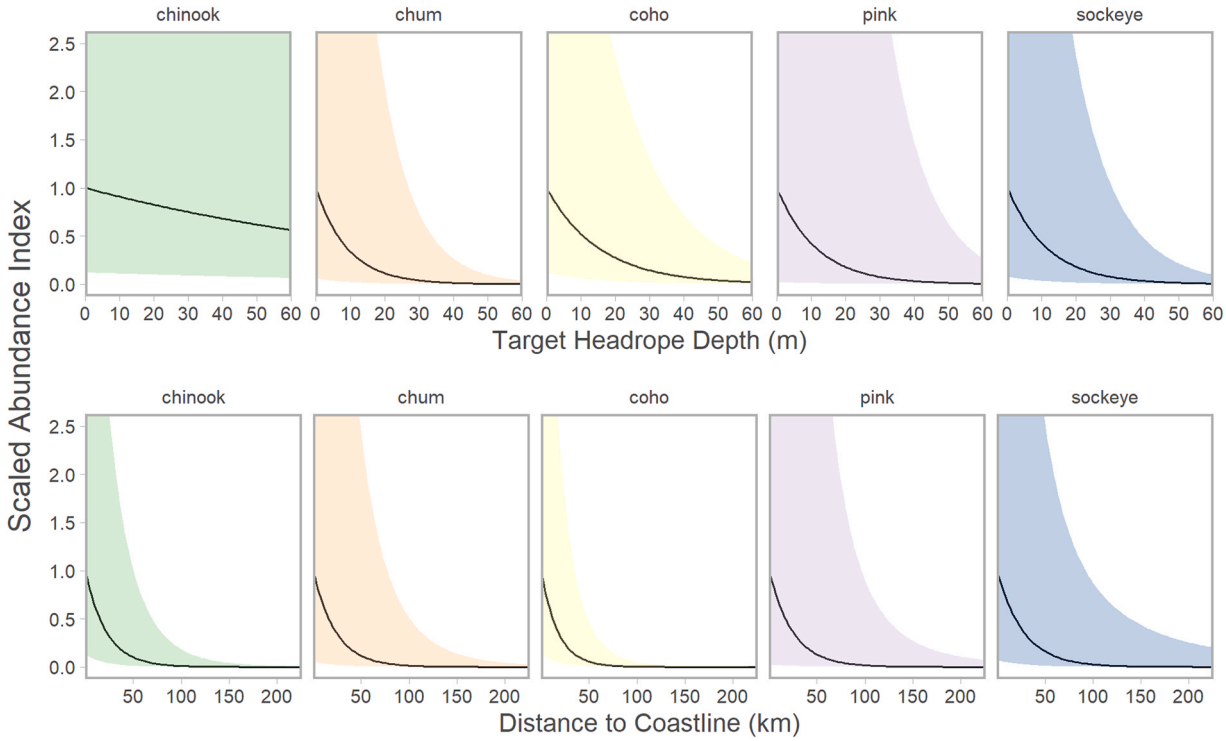


Fig. 4. Changes in predicted scaled abundance as a function of headrope depth (top) and distance to shoreline (bottom). Predictions are conditional effects (covariates fixed to mean or reference values: HSS for survey and day for diel effects) with means (lines) and 95% confidence intervals (ribbons), scaled relative to each species' maximum predicted mean. To improve readability upper y-axes have been constrained such that full confidence intervals are not shown.

S7).

We also found evidence of substantial interannual variability in seasonal distributions, represented by $\epsilon_{s,t}$ (Figures S6, S8, S9). In certain cases model predictions represented well known cycles in abundance (e.g., odd-even year summer abundance of pink salmon in Queen Charlotte Strait; Fig. 6), while in others they appeared to capture short-lived, but dramatic changes in the spatial extent of hotspots (e.g., coho salmon 2009 summer distributions and 2014 fall distributions; Figs. 6, S10).

The predicted abundance of chum, pink, and sockeye salmon was typically greater than that of coho or Chinook salmon. However the abundance of each species also varied interannually, often by multiple orders of magnitude (Figs. 7, S11). Chinook, pink, and sockeye salmon indices showed below average abundance during summer surveys in several recent years (Figure S11), but fall surveys have been relatively

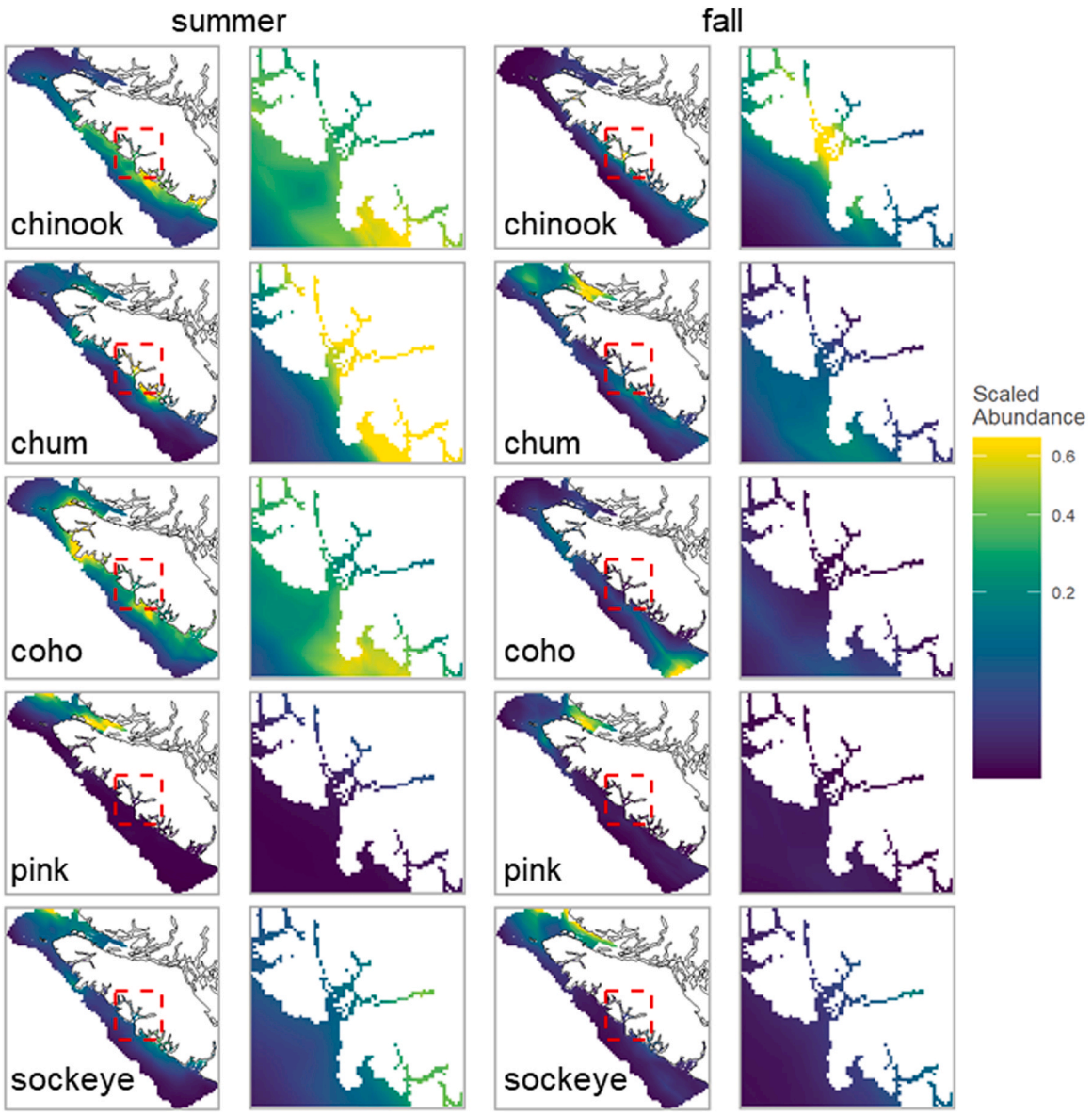


Fig. 5. Predicted median species-specific abundance during summer and fall survey periods. First and third columns show the primary study area, while second and fourth columns focus on inlets with seasonally high abundance. Red dashed polygon represents location of focal inlets. Colors represent the median among year prediction for each cell. Predictions are scaled relative to the species-specific maximum prediction for each season. Yellow cells represent locations where predictions exceeded the 99th percentile of predictions (approximately 60% of the species-specific maximum). Predictions represent conditional effects (covariates fixed to mean or reference values: HSS for survey and day for diel effects).

stationary.

Model comparison using AIC supported including diel and survey design covariates except for Chinook (simpler model supported) and pink salmon (equivalent support for both models; Table S2). Consistent with differences in effect size estimates (Fig. 3) and AIC, abundance indices for some species were sensitive to survey design, while others were not. Indices from both models were similar before changes in survey design occurred (i.e., 2017). After that year, indices of abundance from the model including survey effects were higher for chum, coho, and sockeye salmon (Figure S12). Additionally, indices of abundance from the survey effect model had greater uncertainty for all species (Figure S12), presumably because these models propagated uncertainty from estimating additional fixed effects. Naive models, which excluded parameters for diel and survey design effects, estimated stronger declines in abundance for chum, coho, and sockeye salmon than standard models, with both coho and sockeye salmon showing significant negative trends (Fig. 8). Chinook and pink salmon trends were less sensitive to model structure.

While simulations indicated spatiotemporal models were able to reliably recover the majority of parameter estimates, distance to shore effects and season-specific intercepts were moderately biased relative to true values for some species (Online Supplement). Nevertheless, indices of abundance estimated using the simulated data and associated model fits were unbiased relative to the "true" index (Online Supplement). Thus the models were most sensitive to partitioning variance among spatiotemporal fixed effects (distance to coastline and season-specific mean abundance) and Gaussian random fields, while other fixed effects and interannual patterns in total abundance could be reliably recovered.

4. Discussion

Failing to account for changes in sampling design can bias or reduce the precision of biomass indices that underpin fisheries management,

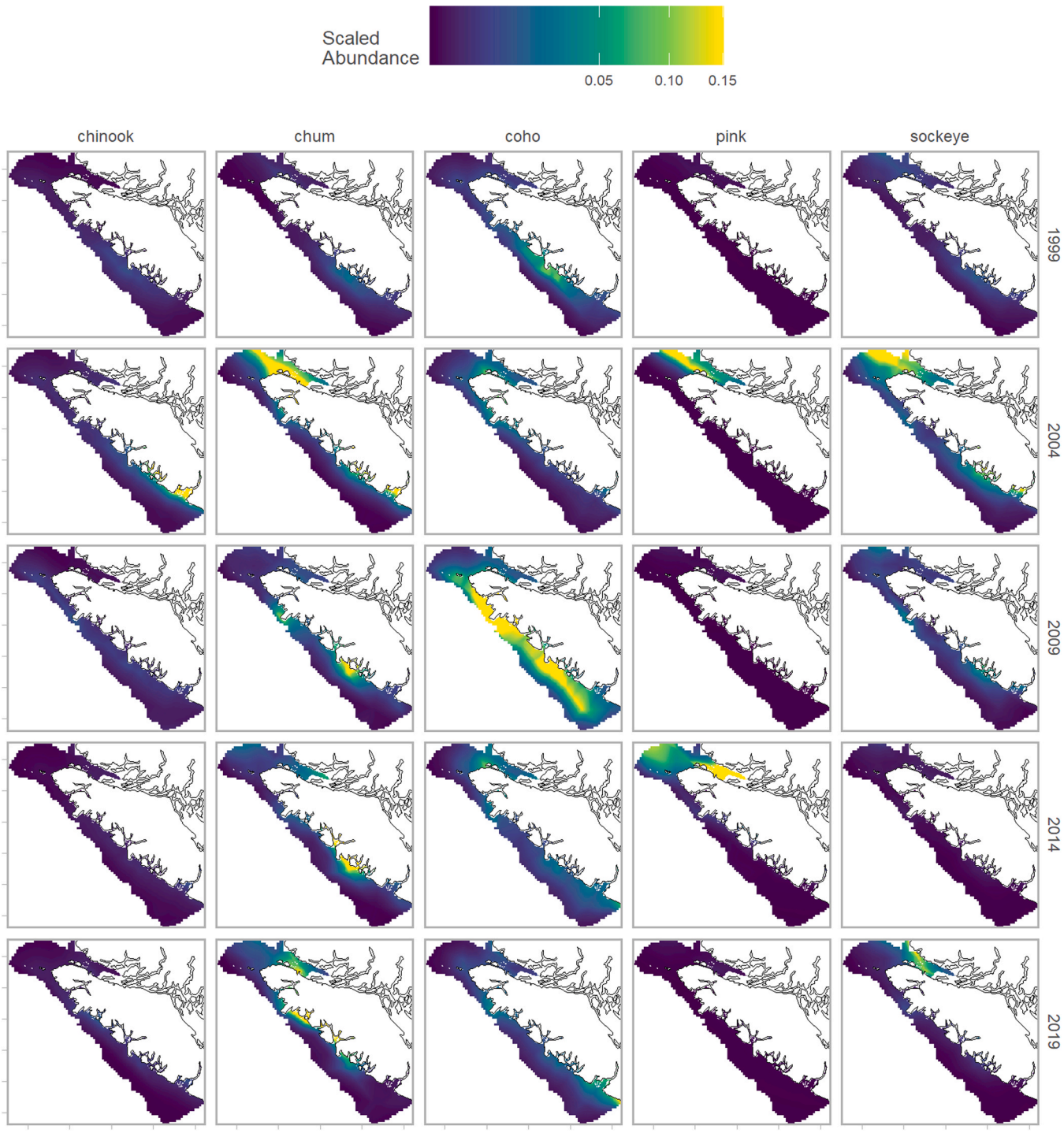


Fig. 6. Predicted species-specific abundance during the summer survey period for a subset of years. Predictions are scaled relative to the species-specific maximum prediction across all seasons. Yellow cells represent locations where predictions exceeded the 99th percentile of predictions. Predictions represent conditional effects (covariates fixed to mean or reference values: HSS for survey and day for diel effects).

regardless of whether the underlying data originate from fisheries or scientific surveys. Spatiotemporal models provide a means of incorporating data collected outside a survey's standard domain, but which still provide information on the processes shaping population abundance. Here we used generalized linear models with Gaussian random fields to estimate annual indices of juvenile Pacific salmon abundance along the continental shelf of British Columbia, while accounting for seasonal variation in spatial distribution, as well as changes in sampling methods. Though juvenile salmon distributions varied at both seasonal and interannual scales, seasonal variability was greater for all species, emphasizing the sensitivity of abundance estimates to juvenile migration phenology. We also found that indices of abundance that did not account for changing survey methods were more likely to show declines in abundance over the previous 12 years for some Pacific salmon species.

4.1. *Survey effects*

We accounted for variation in sampling methods by estimating fixed

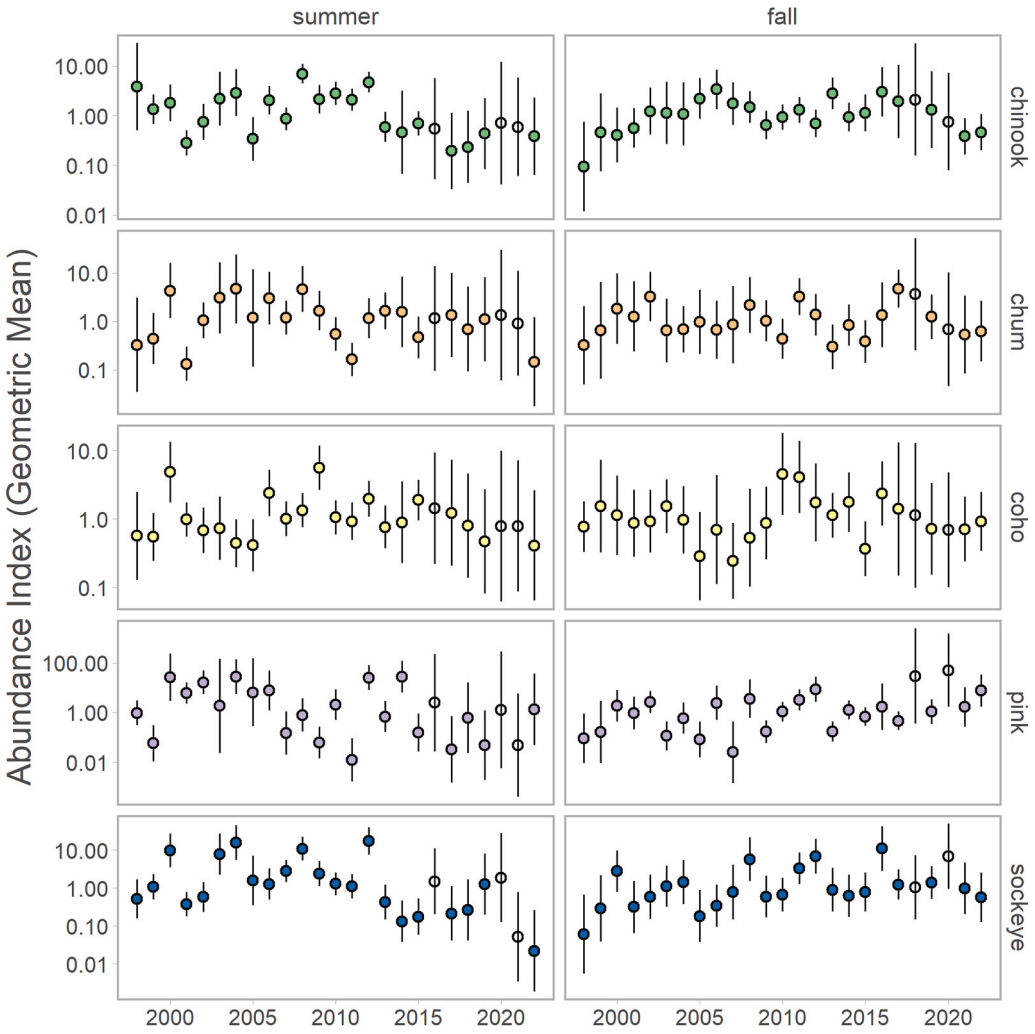


Fig. 7. Model predicted index of abundance during summer and fall. The abundance index represents the predicted number of individuals within the survey domain and is scaled by the geometric mean (e.g., a prediction of 0.1 (10) is an order of magnitude smaller (larger) than the time series geometric mean). Predictions were made assuming a constant HSS survey design, diel sampling, and zero meter headrope depth. Predicted abundance is shown for years with (filled circles) and without (open circles) full survey coverage. Whiskers represent 95% confidence intervals and are only shown for years with full survey coverage to improve readability. Note y-axis tick marks are log distributed.

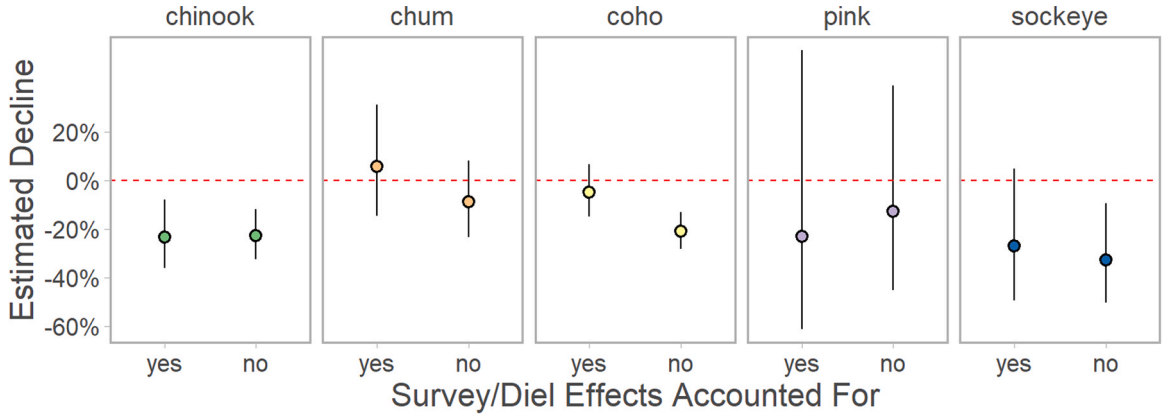


Fig. 8. Estimated percent decline in abundance during summer surveys between 2010 and 2022 from models that did or did not include survey design and diel effects.

effects for distance from shore, diel vs. nocturnal tows, headrope depth, and overall survey design. Distance from shore was a particularly strong driver of abundance among juvenile Pacific salmon. The abundance of all species declined rapidly at distances greater than 15 km from the coastline, which is consistent with previous evidence that juvenile salmon abundance (or catchability) is low off the continental shelf

(Welch et al., 2000; Fisher et al., 2007), and highlights the importance of nearshore and shelf habitats throughout the first year of ocean residence.

Catches of juvenile salmon did not increase with nocturnal sampling. The mean abundance of chum, coho, and sockeye salmon was significantly lower for nighttime tows; pink salmon abundance was lower for nighttime tows, but with greater uncertainty; and Chinook salmon abundance was similar regardless of diel timing. This was counter to our expectations given both salmon competitors and prey (e.g., Pacific herring (*Clupea pallasii*), euphausiids) exhibit diel vertical migrations (Hay and McCarter, 1996; Lu et al., 2003). Additionally catch rates during trawl sampling often increase at night due to reduced gear avoidance (Walsh, 1991). While there are several studies of diel vertical migrations in Pacific salmon, they have generally focused on older life stages of Chinook salmon and evidence of shallowing at night is mixed (Smith et al., 2015; Arostegui et al., 2017; Courtney et al., 2019; Sabal et al., 2023). Lower nocturnal catch rates among four of the five species may reflect poorly understood ecological processes, such as localized responses to prey, forage fish competitors, or predators, and warrant future investigation.

Trawl-based juvenile salmon surveys generally deploy gear near the surface (Fisher et al., 2007), to maximize sampling efficiency. Indeed, we found that catches were strongly and negatively correlated with headrope depth so that predicted abundance declined by more than 90% at depths greater than 20 meters. Chinook salmon were the only exception to this pattern, showing a weaker, linear decline in abundance with depth, which is consistent with evidence that Chinook salmon have a deeper vertical distribution than other species (Orsi and Wertheimer, 1995). It is possible that deeper distribution of Chinook salmon may contribute to weak diel effects described previously, though the ecological mechanism remains unclear.

Finally, we identified modest effects on catch rate after the transition from the historical survey design based on transects and opportunistic sampling (High Seas Salmon; HSS) to the contemporary summer survey in which fishing stations are selected using a stratified random grid and fished during both day and night (Integrated Pelagic Ecosystem Survey; IPES). Mean catch rates were lower for all species, except pink salmon, with the IPES design; however estimates were uncertain and not significantly different from zero. We found that failing to account for diel sampling and survey design effects when calculating indices of abundance impacted trends in abundance for several species (additional details below).

Without detailed simulation testing it is difficult to quantify the trade-offs associated with each survey design. On the one hand, the stratified, randomized design of the IPES survey may be less biased by hotspots of juvenile salmon abundance. On the other, the current IPES design fishes each station twice per survey (day and night). Ultimately fewer locations are sampled and the precision of abundance estimates declines. Unfortunately a relatively large number of summer surveys have been cancelled in recent years due to vessel availability and COVID pandemic impacts on field programs, limiting the statistical power available to resolve this issue. Regardless, the ability of model-based abundance indices to account for changes in survey methods is a key advantage relative to design-based alternatives (ICES, 2023).

4.2. Seasonal patterns

All Pacific salmon species showed large declines in juvenile abundance during spring (i.e., after the first winter of ocean residence and prior to the next cohort's ocean entry. However, it is unclear to what extent these patterns are a result of dispersal off the continental shelf, changes in catchability, or mortality. While abundance was greater during the summer than the fall, the magnitude of this change ranged from relatively large for sockeye salmon to negligible for pink salmon. Each species also showed greater seasonal variability in spatial distributions than interannual variability, consistent with evidence that seasonal migrations are relatively stable year after year (Tucker et al., 2012).

Seasonal changes in distribution varied among species. For example, Chinook, chum, and coho salmon showed similar declines in abundance between summer and fall survey periods, but species-specific changes in distribution. Chinook salmon distributions shifted from diffuse and shelf-oriented during the summer to concentrated within west coast Vancouver Island inlets during the fall. Chinook salmon seasonal distributions are likely the result of stock-specific migrations. During summer months, juvenile Columbia River yearling populations migrate along the west coast of Vancouver Island en route to northern maturation grounds. During the fall survey period, local westcoast Vancouver Island subyearling stocks have begun to disperse into the survey area where they mix with lower Fraser River and Puget Sound subyearling stocks (Fisher et al., 2007; Trudel et al., 2009; Tucker et al., 2011).

Chum salmon distributions shifted from the west coast to north of Vancouver Island during fall. While chum salmon movements are more poorly resolved than many other species, these patterns are are broadly consistent with a northern migration along the continental shelf (Fisher et al., 2007).

Coho salmon were widely distributed along the continental shelf during the summer, then concentrated in southern portions of the survey domain during the fall. Coho salmon have predominantly yearling freshwater life histories, which may contribute to a similar summer distribution to Chinook salmon (when yearling Chinook salmon populations are most abundant in the area). The southerly fall distribution is unusual compared to other Pacific salmon species; however, coho salmon rear in relatively close proximity to their natal streams (Weitkamp and Neely, 2002) and historically coho salmon catches were high in southwest Vancouver Island (Healey et al., 1990). Thus locations in the southern portion of our survey domain, such as Barkley and Juan de Fuca canyons, may represent important rearing habitats.

Although pink and sockeye salmon showed relatively stable spatial distributions between summer and fall surveys, there are several nuances. First, seasonal declines in abundance were much larger for sockeye than pink salmon, suggesting a greater proportion of pink salmon stocks, or individuals within stocks, rear in southern areas for a relatively longer period. Second, Queen Charlotte Strait showed a relatively strong odd-even year pattern in pink salmon abundance during the summer, but not during the fall. We propose that pink salmon caught in this region during summer months are predominantly from the Fraser River, while during the fall a greater proportion of east coast Vancouver Island and Puget Sound populations are present. The Fraser River is dominated by odd-year runs of pink salmon, while the other two regions have relatively large contributions from both cycle lines.

4.3. Interannual variability

Indices of abundance for each species varied by orders of magnitude over the 25-year time series. Interannual variability was particularly strong for pink and sockeye salmon, consistent with well resolved cycles in abundance for several Fraser River populations (Groot and Margolis, 1991). Juvenile salmon species also exhibited differences in the relative stability of spatial distributions among years. Chinook and sockeye salmon distributions were typically concentrated in approximately the same location, year after year. Conversely, chum salmon showed greater variability with high abundance on the west coast of Vancouver Island in certain years and in Queen Charlotte Strait in others. Stable distributions may reflect relatively less individual plasticity in habitat selection. Alternatively, if distributions are strongly correlated with stock identity then interannual variability in distribution may be largely driven by interannual changes in stock-specific abundance.

Interannual variability in both mean abundance and spatial distribution may be driven by interacting processes. Large differences in abundance can arise directly due to variability in the number of juveniles entering the marine environment. Knock-on density dependent

effects during marine residence could further depress abundance or alter distributions (Levin et al., 2001; Freshwater et al., 2017). Additionally small pelagic species are often patchily distributed, resulting in challenges separating process from observation error when modeling abundance or distributions (Boyd et al., 2015). These issues are exacerbated in juvenile Pacific salmon where variability in ocean entry phenology, marine growth, or ecological conditions can moderate migration behaviour (Wells et al., 2016; Freshwater et al., 2019), and therefore the proportion of each population that is vulnerable to sampling in a given year.

Trends in juvenile Pacific salmon abundance indices during the summer survey differed among species in two ways. First, using a model that incorporated survey effects, Chinook salmon showed evidence of a significant decline in summer abundance over the previous 12 years. Sockeye salmon showed a decline, though with considerable uncertainty, while the remaining species' dynamics were stationary. Second, estimated declines in some species, but not others, were sensitive to whether diel and survey design effects were accounted for. Negative trends in abundance steepened for chum, coho, and sockeye salmon when a naive model that excluded diel and survey effects was used to estimate abundance.

4.4. Caveats

Our analysis has several important limitations. First, we did not identify stock-specific patterns in spatial distribution or abundance within a given species. Pooling stocks is likely to be particularly problematic for Chinook salmon, which include a wide range of populations that can exhibit substantial variability in migration patterns (Weitkamp, 2010; Tucker et al., 2011; Freshwater et al., 2021). Our model is relatively robust to coarse differences among stocks because it accounts for seasonal and annual changes in distribution, as well as abundance. However, future analyses that incorporate stock composition estimates will be necessary to resolve stock-specific distributions. Such analyses will be particularly important for improving our understanding of the marine ecology of chum, coho, and pink salmon, which have been understudied relative to Chinook and sockeye salmon.

Second, our estimates of survey design effects are potentially confounded with interannual variation because it was not possible to complete both surveys simultaneously over multiple years. We constrained survey design effects to account for this limitation by modeling multiple seasons simultaneously (the fall survey design never changed) and by incorporating temporal autocorrelation in catch rates. These constraints appeared to be effective since simulation testing indicated survey effect parameter estimates could be reliably recovered. Nevertheless, there is additional variability in survey methods that we were unable to account for. Year-to-year variation in factors such as tow speed, vessel, and captain may impact catch rates, but were not addressed here due to data limitations and more complex models failing to converge. Such findings emphasize the strengths and limitations of model-based approaches to estimating abundance—while more flexible than design-based approaches, they still require adequate data to disentangle interacting processes (ICES, 2023).

Third, estimates of abundance collected using midwater trawl surveys have ecological caveats. Headrope depth is an imperfect proxy for vertical distributions because it integrates over the entire mouth opening (median 13 m in our dataset) and it is possible for fish to be captured during deployment or retrieval (i.e., between the surface and target headrope depth). Given the size of midwater trawl vessels, it is also not possible to fish in shallow depths. As a result juveniles are not vulnerable to the survey until they begin to migrate away from nearshore environments and our models are unlikely to accurately predict abundance within the 50 m isobath.

Fourth, we did not directly incorporate environmental covariates into our model. Assessment models are often used to estimate interannual abundance, then environmental drivers of interannual variation are evaluated post-hoc (e.g., Farley et al., 2024). There may be benefits to an integrated approach, however, when environmental conditions are predicted to influence catchability rather than abundance. For example, indices of abundance for Chinook salmon in the Bering Sea account for changes in habitat availability associated with mixed layer depth (Murphy et al., 2017). Since southern British Columbia does not have oceanographic features analogous to the Bering Sea cold pool (Thomson, 1981), mixed layer depth is less likely to impact catchability in our study domain. Nevertheless, future work would benefit from considering whether alternative environmental covariates may moderate catchability.

4.5. Conclusions

The spatiotemporal models we presented here could be adapted to meet several management needs. Increasingly, fisheries agencies are tasked with contributing to spatial planning processes for protected areas (e.g., DFO, 2021). The models presented here are currently being adjusted to emphasize seasonal distributions, rather than interannual abundance, and fit to stock-specific catch data, derived from genetic stock identification sampling, to generate high-resolution predictions of stock-specific distributions. Similarly, while our index captures broad patterns in juvenile abundance, such as four-year cycles in the abundance of Fraser River sockeye salmon (COSEWIC, 2017), our goal was to evaluate the potential impact of survey design and provide *species-specific* indices of abundance. The stock-specific indices relevant to assessments and return forecasts could be developed by incorporating stock-specific catch data, again based on genetic sampling, into the current modeling framework. Such outputs could also be used to identify drivers of productivity by associating stock-specific indices of marine juvenile abundance with recruitment and, for data-rich populations, estimates of freshwater juvenile abundance. Finally, environmental covariates could be incorporated to quantify linkages between salmon distributions or abundance and ecological processes, such as upwelling intensity or prey availability.

Fisheries-independent surveys provide information critical to traditional stock assessments and ecosystem-based management. Yet changes in survey design are inevitable and can bias or reduce the precision of survey data for migratory species, which are particularly sensitive to the timing and location of sampling. We developed a spatiotemporal model with sufficient flexibility to account for seasonal and interannual changes in both the distribution and abundance of migratory species, as well as changes in survey design. Our results provide an example of the sensitivity of indices of abundance to changes in survey methods and provide a framework for future investigations into mechanisms regulating Pacific salmon recruitment.

CRediT authorship contribution statement

Jackie King: Writing – review & editing, Supervision, Resources, Data curation, Conceptualization. **Sean C. Anderson:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Cameron Freshwater:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data and code underpinning these analyses are available from https://github.com/pbs-assess/juvenile-salmon-index and have been archived with Zenodo at https://doi.org/10.5281/zenodo.11210744.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.fishres.2024.107063.

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